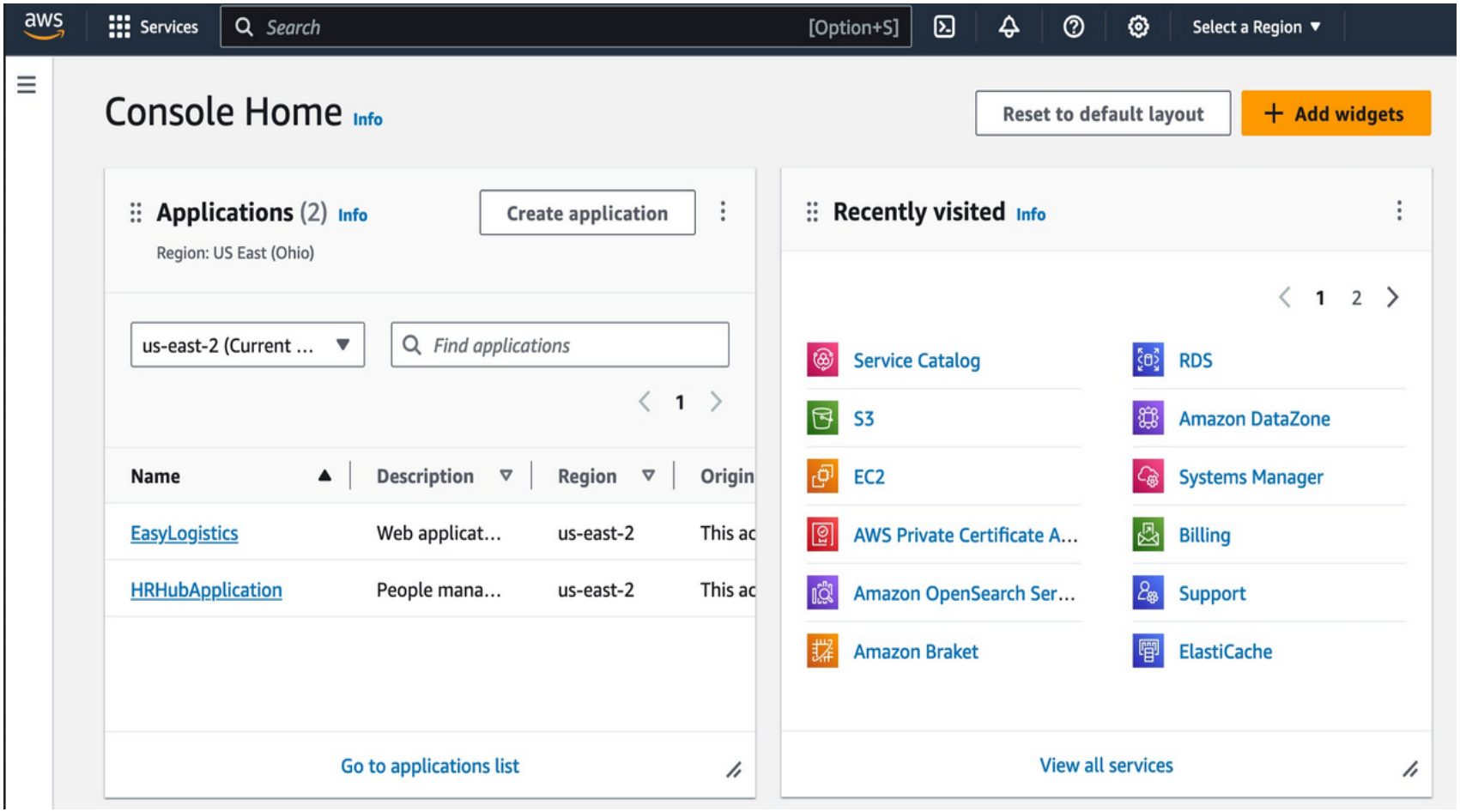
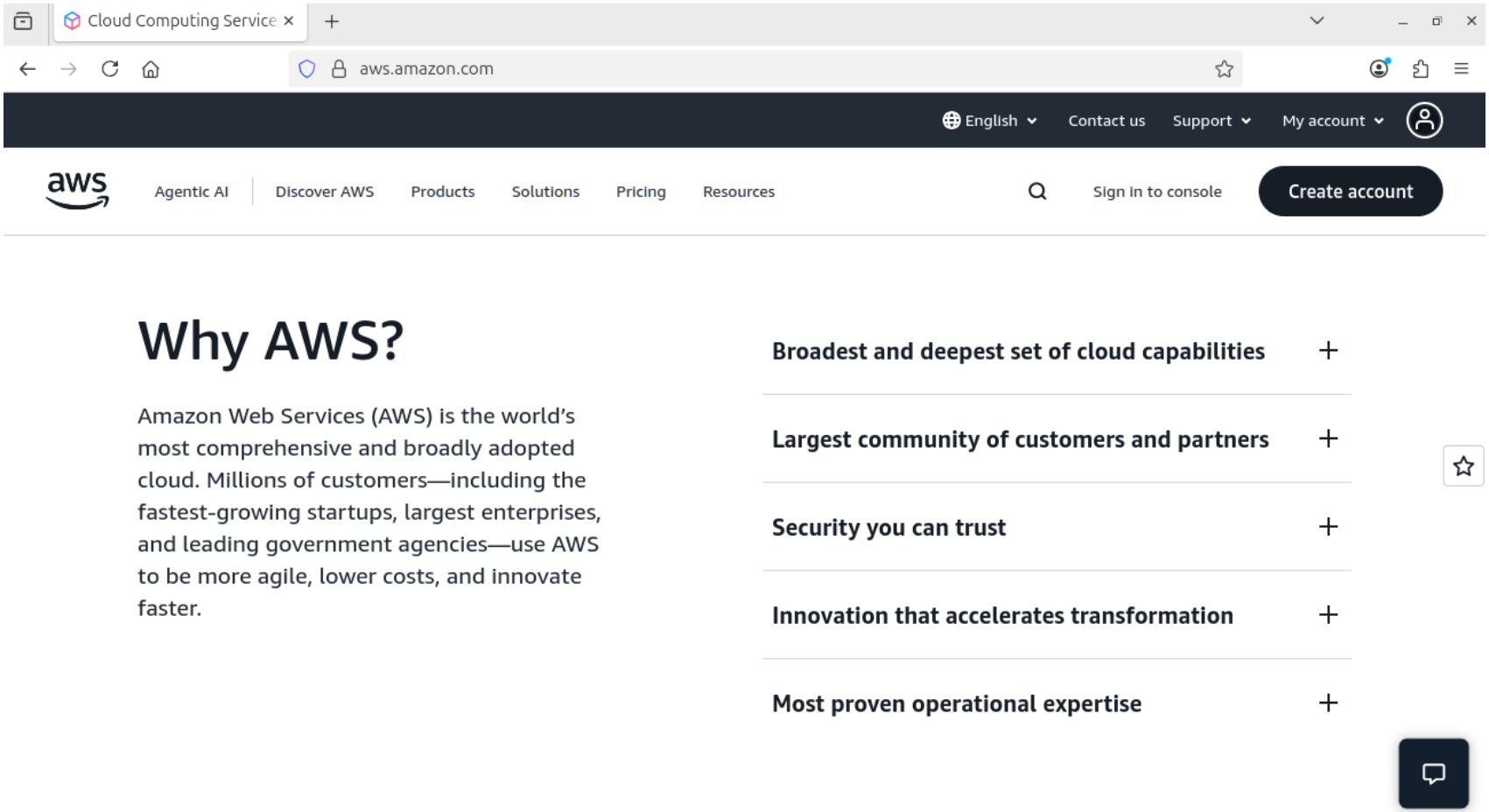
IT Workshop II (Cloud Computing): Introduction, Cloud Concepts and Technologies

July 30, 2026

Amazon Web Services (AWS)



Amazon Web Services (AWS)



Python: Classes

Python is an Object-Oriented Programming (OOP) language. Python provides all the standard features of Object Oriented Programming such as classes, class variables, class methods, inheritance, function overloading, and operator overloading.

```
# Examples of a class
class Student:
        studentCount = 0

        def __init__(self, name, id):
                print "Constructor called"
                self.name = name
                self.id = id
                Student.studentCount = Student.studentCount + 1
                self.grades={}

        def __del__(self):
                print "Destructor called"

        def getStudentCount(self):
                return Student.studentCount

        def addGrade(self,key,value):
                self.grades[key]=value
                def getGrade(self,key):
                return self.grades[key]

        def printGrades(self):
                for key in self.grades:
                        print key + ": " + self.grades[key]
```

```
>>>s = Student('Steve','98928')
Constructor called

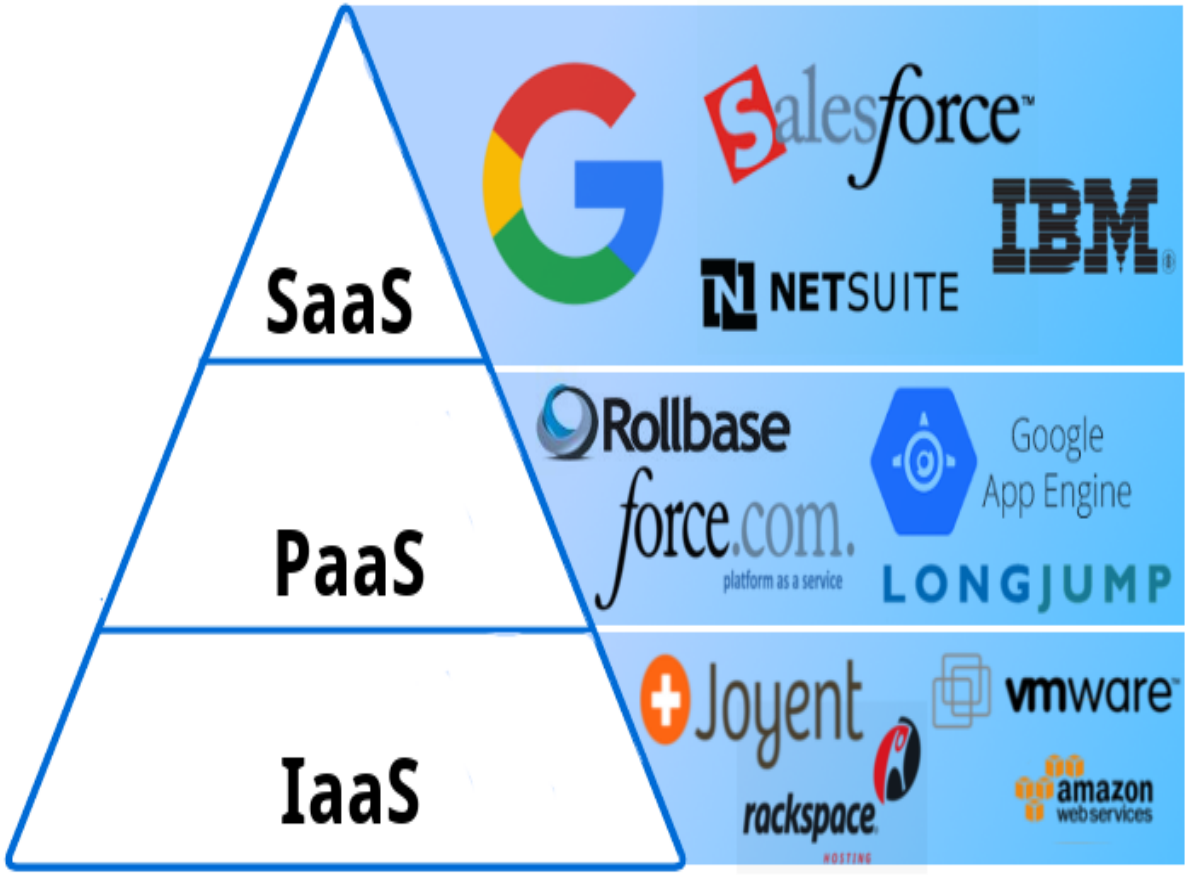
>>>s.addGrade('Math','90')
>>>s.addGrade('Physics','85')
>>>s.printGrades()
Physics: 85
Math: 90

>>>mathgrade = s.getGrade('Math')
>>>print mathgrade
90

>>>count = s.getStudentCount()
>>>print count
1

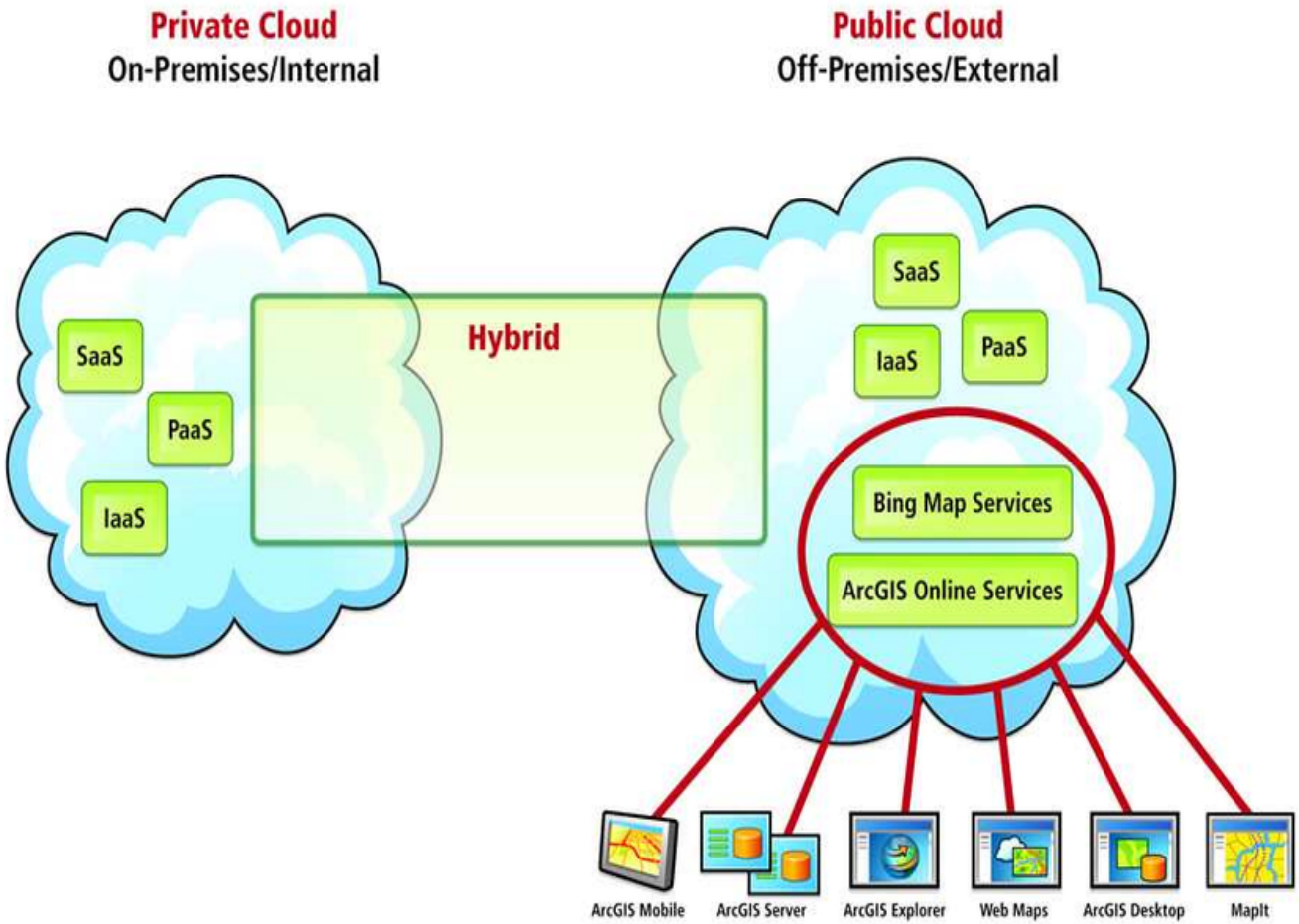
>>>del s
Destructor called
```

SaaS, PaaS, IaaS



Source: Internet

Cloud Computing Deployment Models



Source: Internet

Cloud Computing Applications

- Banking and Financial Apps
- E-Commerce Apps
- Social Networking
- Healthcare Systems
- Intelligent Transportation Systems
- Mobile Communications

Virtualization

- Virtualization refers to the partitioning the resources of a physical system (such as computing, storage, network and memory) into multiple virtual resources.
- Virtualization is the key enabling technology of cloud computing and allows pooling of resources.
- In cloud computing, resources are pooled to serve multiple users using multi-tenancy.
- Multi-tenancy aspects of the cloud allow multiple users to be served by the same physical hardware.

The End